

Question

Organic extractant HA can extract Ce^{3+} from aqueous solution into the organic phase in the form of $\text{Ce}(\text{A})_x(\text{HA})_y$. HA is almost insoluble in water, but exists in the organic phase only as a dimer of H_2A_2 . In order to determine the specific form of this cerium complex in the organic phase, the following experiment was carried out: A tracer amount of $^{144}\text{Ce}^{3+}$ was dissolved in water and acidified with a small amount of nitric acid to prepare an aqueous phase containing $^{144}\text{Ce}^{3+}$; 2.50 mL each of the above aqueous phase and the organic phase containing HA were transferred and thoroughly mixed and extracted at room temperature. After equilibrium, 1.00 mL each of the aqueous phase and the organic phase were taken, and the ratio of their radioactivity intensities was measured; another portion of the equilibrated aqueous phase was taken to measure its pH value. When the initial concentration of H_2A_2 is $c_0(\text{H}_2\text{A}_2) = 0.240 \text{ mol/L}$, the ratio of the radioactivity intensities of the organic phase to the aqueous phase at equilibrium is measured to be 2929 : 416, and the aqueous phase $[\text{H}^+] = 3.39 \times 10^{-2} \text{ mol/L}$; when $c_0(\text{H}_2\text{A}_2) = 0.120 \text{ mol/L}$, the ratio of the radioactivity intensities of the organic phase to the aqueous phase at equilibrium is measured to be 1542 : 1682, and the aqueous phase $[\text{H}^+] = 3.42 \times 10^{-2} \text{ mol/L}$.

The following statements are made:

1. The sum of x and y is an even number (treat x and y as integers).
2. The sum of the coefficients on both sides of the simplified and balanced extraction reaction equation are not equal.
3. The equilibrium constant of the extraction reaction is smaller than 0.023.
4. The equilibrium constant of the extraction reaction is smaller than 0.015.
5. The equilibrium constant should be taken to 2 significant figures.

The hydrolysis of Ce^{3+} and the complexation of Ce^{3+} with NO_3^- are not considered in the calculation of the equilibrium constant. What is the sum of the correct serial numbers?

A. 1

B. 2

C. 3

D. 4

E. 5

F. 7

G. 9

H. 10

Answer

Correct Answer: **D**

Detailed Explanation

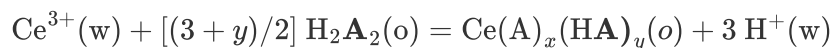
Since the extracted substance $\text{Ce}(\mathbf{A})_x(\text{HA})_y$ is electrically neutral, $x = 3$;

CHECKPOINT

1 PTS

$$x = 3$$

The extraction reaction equation is:



CHECKPOINT

2 PTS

The extraction reaction equation is: $\text{Ce}^{3+}(\text{w}) + [(3 + y)/2] \text{H}_2\mathbf{A}_2(\text{o}) = \text{Ce}(\mathbf{A})_x(\text{HA})_y(\text{o}) + 3 \text{H}^+(\text{w})$

$$K = \frac{2929 \times [3.39 \times 10^{-2}]^3}{416 \times 0.240^{(3+y)/2}} = \frac{1542 \times [3.42 \times 10^{-2}]^3}{1682 \times 0.120^{(3+y)/2}}$$

CHECKPOINT

2 PTS

Equilibrium constant calculation equation: $K = \frac{2929 \times [3.39 \times 10^{-2}]^3}{416 \times 0.240^{(3+y)/2}} = \frac{1542 \times [3.42 \times 10^{-2}]^3}{1682 \times 0.120^{(3+y)/2}}$

Solving the above equation, we obtain: $y = 2.9 \approx 3$; the value of K is between $1.50 \times 10^{-2} \sim 2.30 \times 10^{-2}$, retaining 3 significant figures.

CHECKPOINT

1 PTS

$$y = 2.9 \approx 3$$

CHECKPOINT

1 PTS

The value of K is between $1.50 \times 10^{-2} \sim 2.30 \times 10^{-2}$

$x = 3, y = 3$, so the sum is equal to 6, which is an even number. Statement 1 is correct.

The equilibrium constant of the extraction reaction is smaller than 0.023 and larger than 0.015, statement 3 is correct, and 4 is wrong.

Therefore, statements 1 and 3 are correct, choose option D.